

ShopEZ E-Commerce Shopping Platform

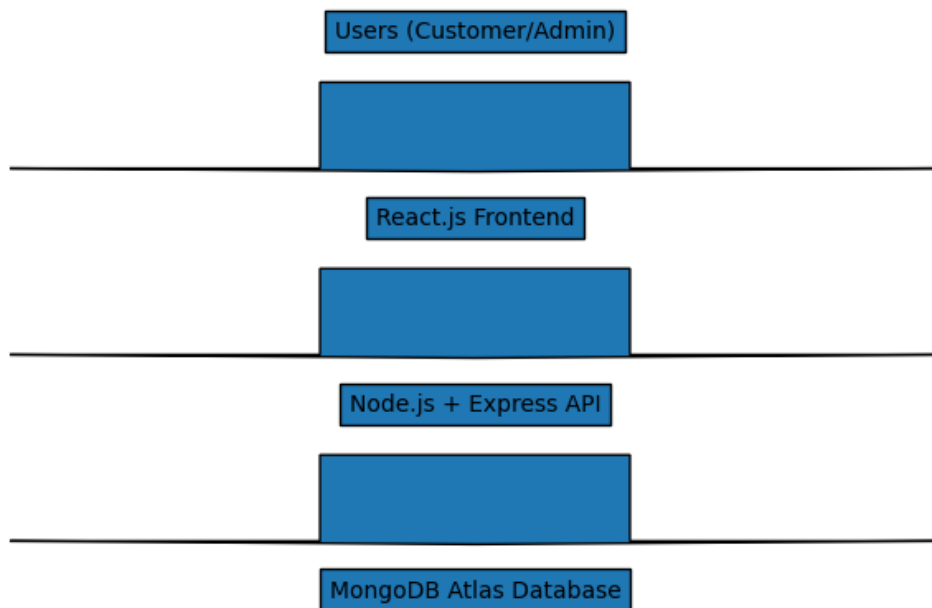
- Solution Architecture

Professional Architecture Document with Diagrams

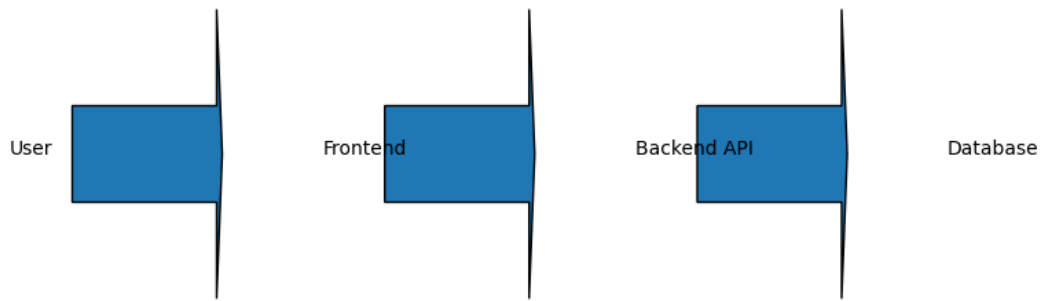
Architecture Overview

ShopEZ follows a scalable 3-tier MERN architecture consisting of a React.js frontend, Node.js/Express backend services, and MongoDB Atlas database. The platform supports authentication, product management, shopping cart, orders, payments, analytics, and administration.

System Architecture Diagram



Data Flow Diagram



Architecture Layers

- Presentation Layer: React.js, Bootstrap, React Router
- Application Layer: Node.js, Express.js, JWT Authentication
- Data Layer: MongoDB Atlas and Mongoose
- Integration Layer: Payment Gateway, Cloudinary, Email Services
- Security Layer: JWT, bcrypt, HTTPS, RBAC